

JKBOSE 2026

CLASS 11TH

#XUV 

REDOX

REACTIONS

LECTURE-02 CHEMISTRY

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Bhaichara
ON
TOP.



JKBose Wallah

Topics To Covered

- 1** Calculation of Oxidation Number
- 2** Types of Redox Reactions

REVISION

Oxidation

Old concept

- 1) Addition of O or more EN. →
- 2) Removal of H or more electropositive →

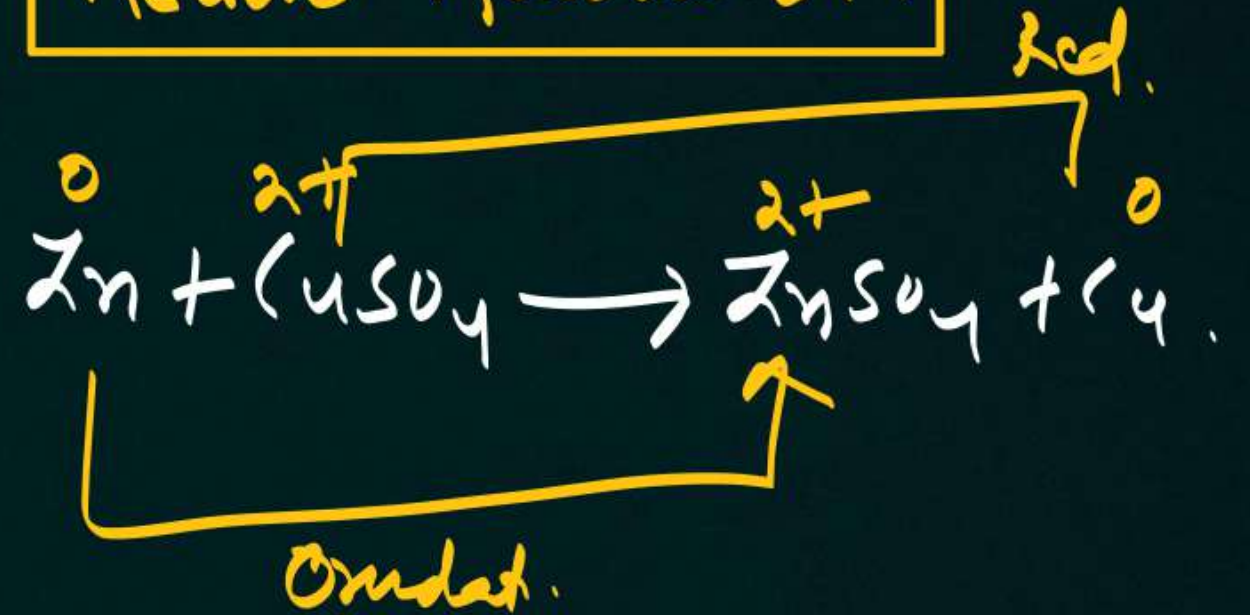
Modern concept

Loss of e^- →

Reduction

Old concept

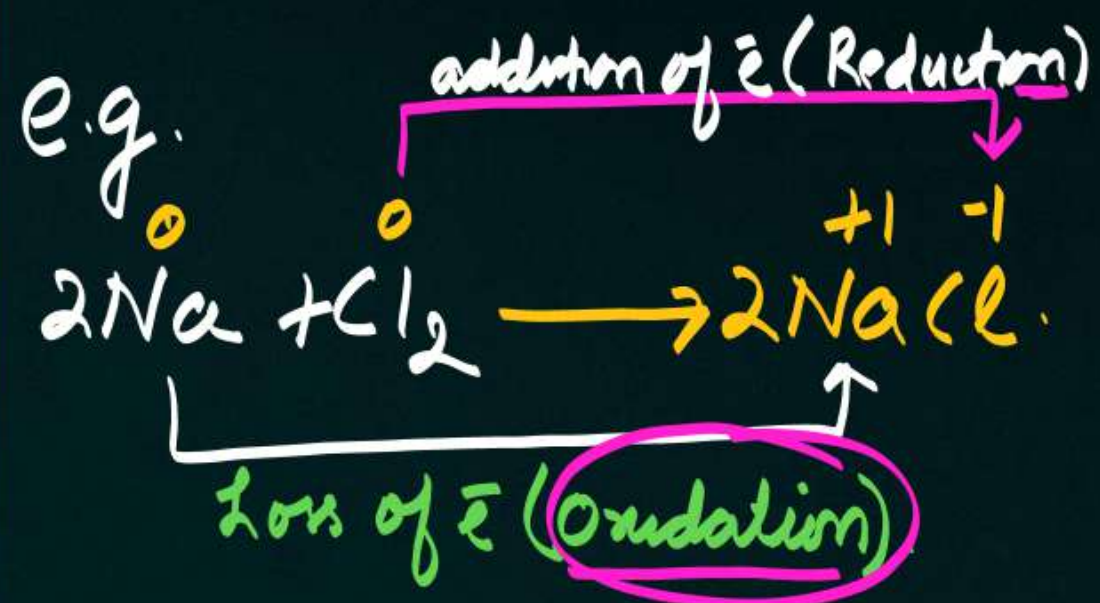
Redox Reaction :-



Oxidising and Reducing Agent

Oxidising Agent or Oxidant $\rightarrow$ Substance that gains electron or undergoes reduction during a reaction is called oxidising agent.

Reducing agent or Reductant $\rightarrow$ Substance that loses electrons or undergo oxidation during a reaction is called reducing agent.

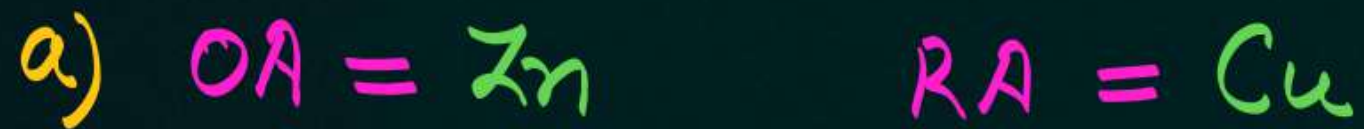
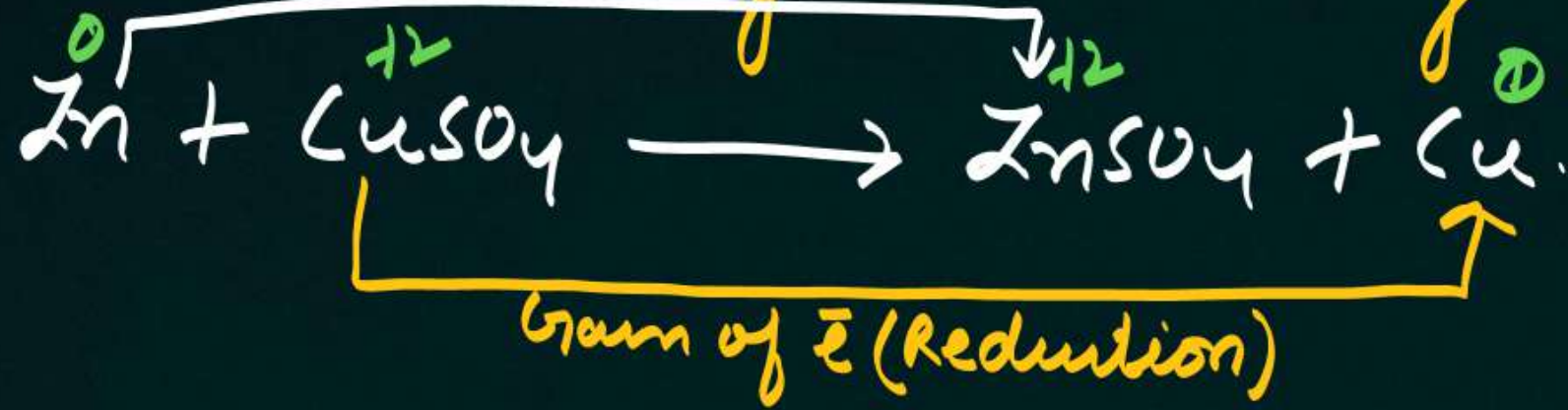


In the above reaction

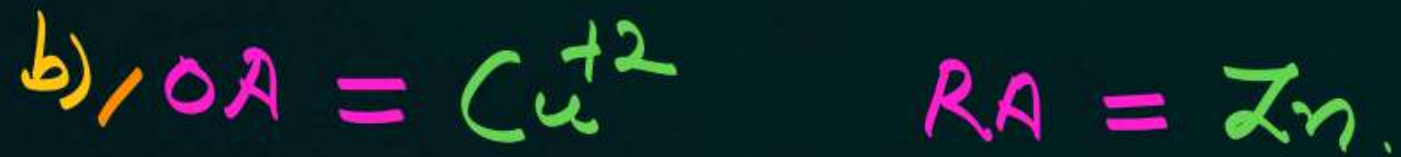
Cl_2 acts as oxidising Agent
+

Na acts as Reducing agent.

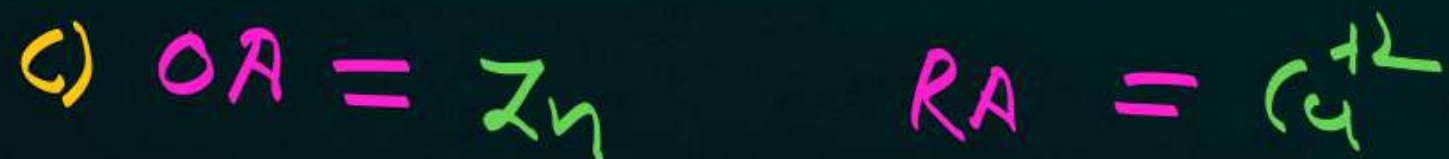
Q. Name the oxidising and reducing agent



Reduction $\rightarrow$ OA ✓



Oxidation $\rightarrow$ RA ✓



SPECIAL CASES.

1) C_3O_2 (Carbon suboxide).



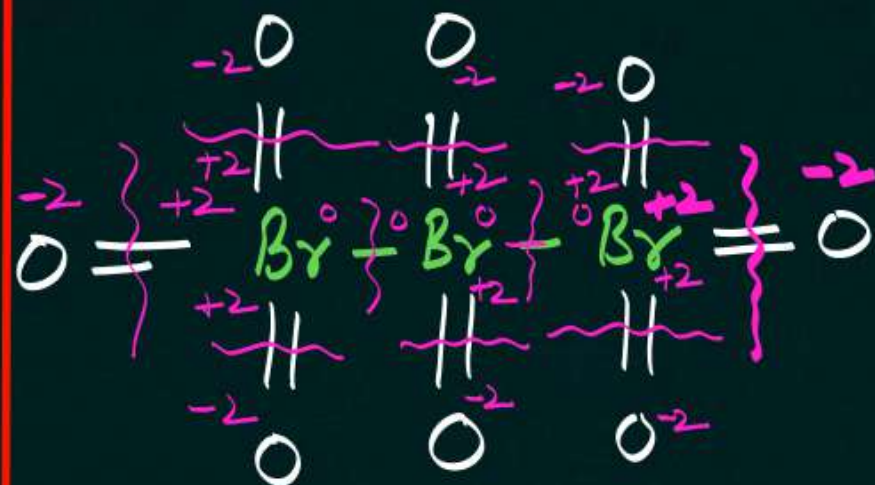
Two $O = -2$.

Two $C = +2$.

One $C = 0$.

More EN = -ve
less EN = +ve

2) Br_3O_8 (Tribromo octaoxide)



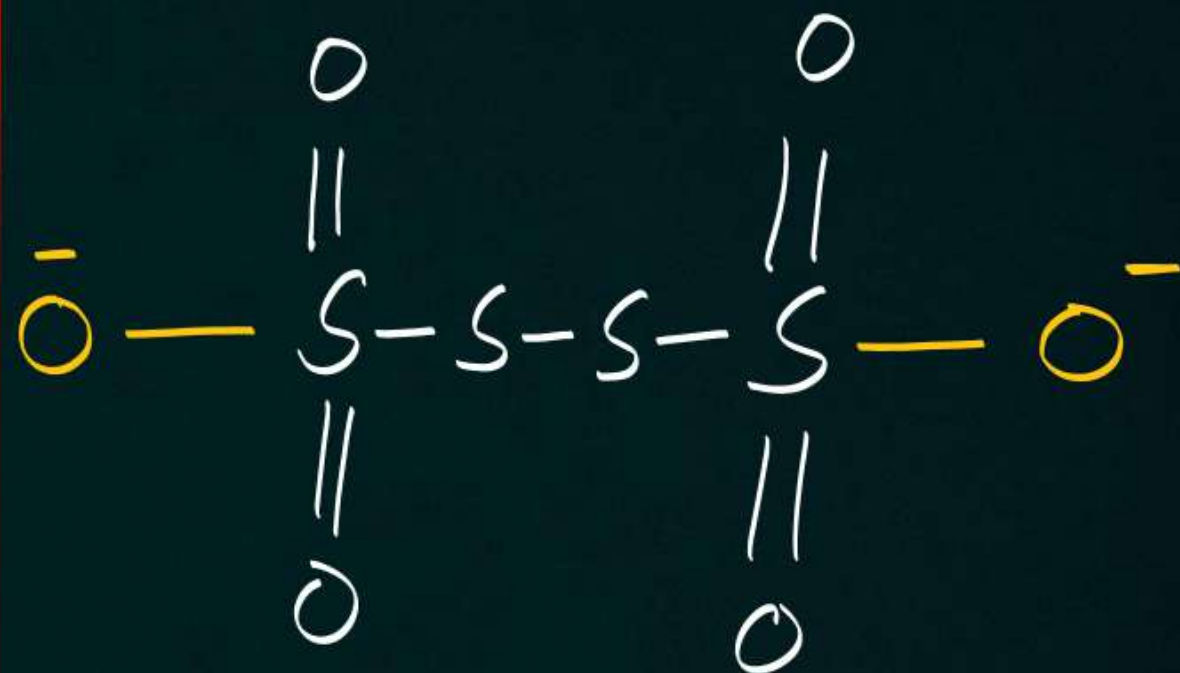
Eight $O = -2$.

Two $Br = +6$

One $Br = +4$.

$O > Br$
EN EN

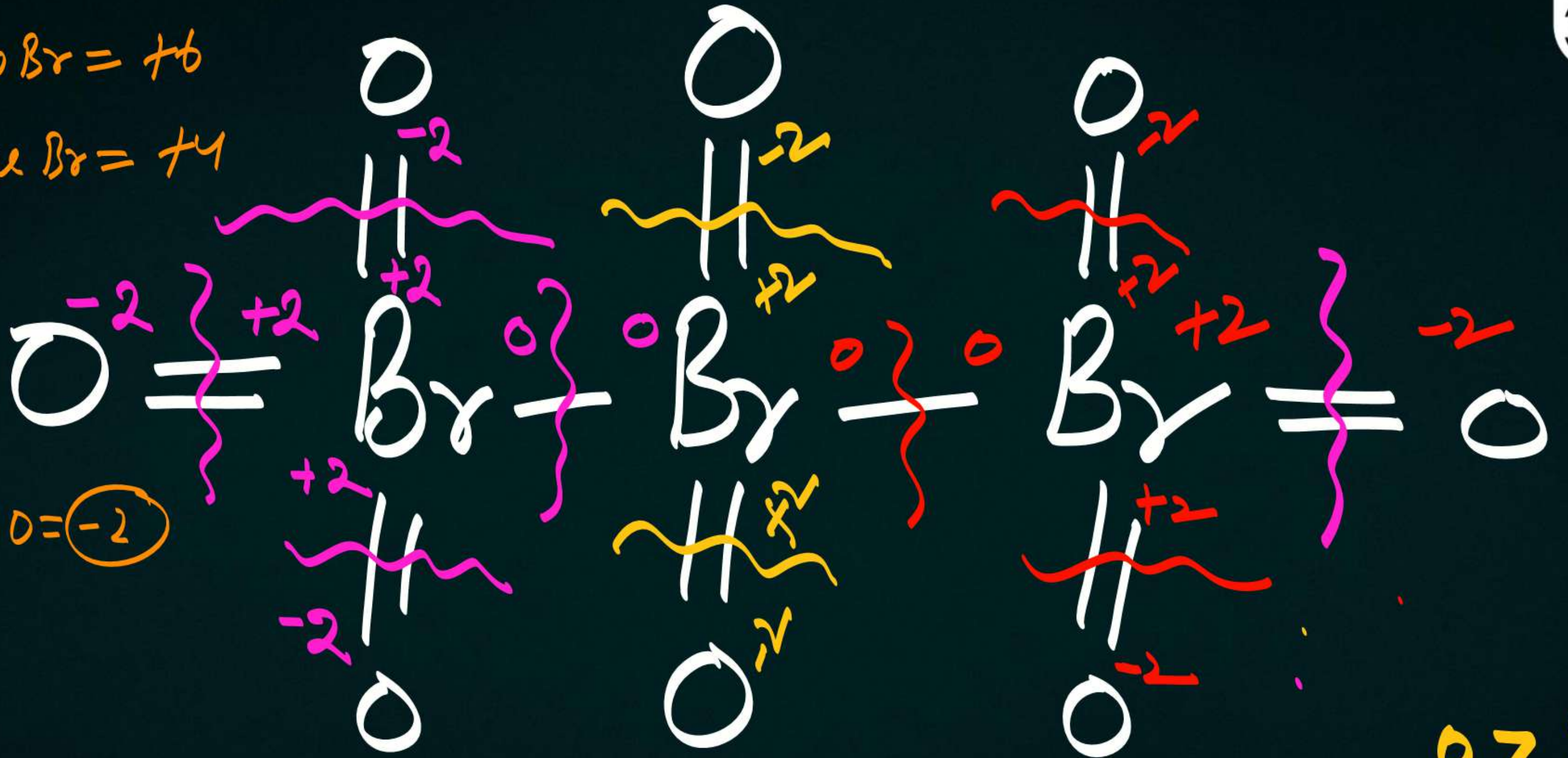
3) $S_4O_6^{2-}$ (Tetrathionate ion)



Two Br = +6

One Br = +4

All O = -2



O 7 Br
EN EN

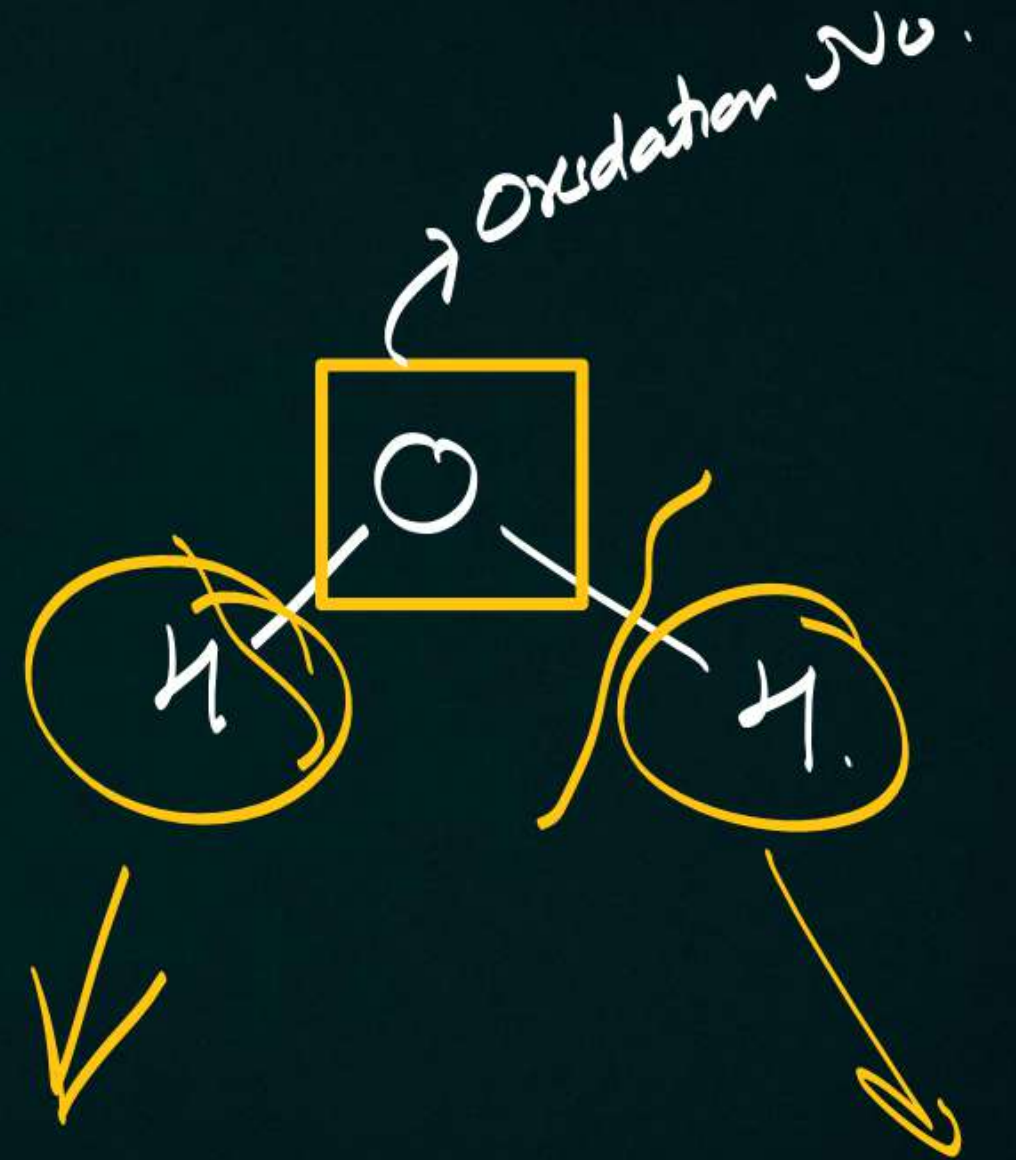


Calculation of Oxidation Number

1. **Free/elementary state:** Oxidation number = 0 (e.g., H_2 , O_2 , O_3 , Fe, C (diamond), Br_2 , P_4 , S_8).
2. **Monoatomic ion:** Oxidation number = charge on ion
(e.g., $\text{K}^+ = +1$, $\text{Ca}^{2+} = +2$, $\text{Al}^{3+} = +3$, $\text{Cl}^- = -1$)
3. **Metal + Non-metal compounds:** Metal = positive, Non-metal = negative
(e.g., $\text{NaCl} \rightarrow \text{Na} = +1$, $\text{Cl} = -1$).
4. **Hydrogen:** Usually +1; in metal hydrides (LiH , NaH , CaH_2 etc.) = -1.
5. **Oxygen:** Usually -2.
 - Peroxides (H_2O_2 , Na_2O_2): -1 each.
 - Superoxides (KO_2 , RbO_2): $-\frac{1}{2}$ each.
 - With fluorine (OF_2 : +2; O_2F_2 : +1 each).
6. **Fluorine:** Always -1 (most electronegative). Other halogens = -1, except when bonded to a more electronegative atom.
7. **Neutral molecules:** Sum of oxidation numbers = 0 (e.g., NH_3).
8. **Polyatomic ions:** Sum of oxidation numbers = net charge on ion (e.g., SO_4^{2-}).

Oxidation Number

Charge present on any atom in a molecule, when all other atoms are removed.



Calculation of Oxidation Number.

Rules

1) Charge present on free atom or elementary state of atom is always zero.



2) Oxidation no. of any ion is equal to the charge present on that ion.



3) In case of compounds in metal and non-metals are bonded. +ve charge is given to metal and -ve charge is given to non-metal. e.g. NaCl, Na will acquire +ve charge. & Cl will acquire -ve charge.

4) Elements of Group I (H, Li, Na, K, Rb, Cs, Fr) always have +1 Oxidation No. except 'H'.

" " Group 2 (Be, Mg, Ca, Sr, Ba, Ra) always " +2 Oxidation No.

5) Hydrogen usually have $+1$ Oxidation Number except when bonded with metal (metal hydride).
then will have -1 charge e.g. $\overset{+1}{K}\overset{-1}{H}$, $\overset{+1}{Na}\overset{-1}{H}$, $\overset{+2}{Ca}\overset{2(-1)}{H_2}$ etc

6) Oxygen usually shows -2 Oxidation No. But in case.

Peroxides e.g. H_2O_2 , Na_2O_2 shows -1 charge.

Superoxides e.g. KO_2 , RbO_2 " $-\frac{1}{2}$ charge.

Fluoride e.g. $O_2F_2 = +1$, $OF_2 = +2$.

7) In case halogens Fluorine always show -1 Oxidation Number. Others will also -1 Oxidation Number except when attached with more electronegative elements.

8) In case of Neutral atom total charge is equal to Zero.

9) In case of ion " " " " " charge present on that ion.

e.g.

$$\text{H}_2\text{O}$$

$$2(+1) + x = 0$$

$$2 + x = 0$$

$$x = -2$$

$$\text{NH}_3$$

$$x + 3(+1) = 0$$

$$x + 3 = 0$$

$$x = -3$$

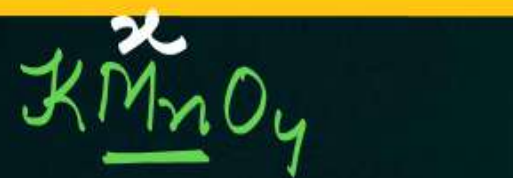
$$(\text{NH}_4)^+$$

$$x + 4(+1) = +1$$

$$x + 4 = +1$$

$$x = 1 - 4$$

$$= -3$$



$$1 + x + 4(-2) = 0$$

$$1 + x - 8 = 0$$

$$x + 1 - 8 = 0$$

$$x - 7 = 0$$

$$x = +7$$



$$2(+1) + 2x + 7(-2) = 0$$

$$2 + 2x - 14 = 0$$

$$2x + 2 - 14 = 0$$

$$2x - 12 = 0$$

$$2x = 12$$

$$x = \frac{12}{2} = +6$$



$$2(+1) + 2x = 0$$

$$2 + 2x = 0$$

$$2x = -2$$

$$x = \frac{-2}{2} = -1$$

QUESTION

Calculate the oxidation number of sulphur in the following molecules/ions:

- (a) H_2S
- (b) H_2SO_3
- (c) SO_4^{2-}
- (d) $\text{Na}_2\text{S}_2\text{O}_3$
- (e) $\text{S}_2\text{O}_7^{2-}$
- (f) H_2SO_4
- (g) $\text{S}_2\text{O}_4^{2-}$

N.W

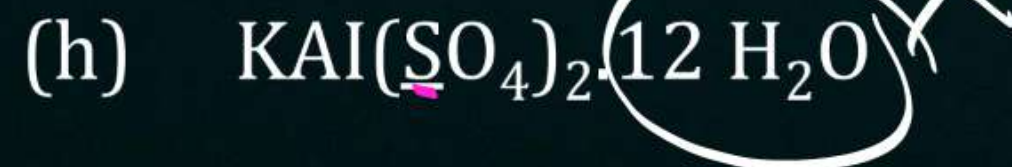
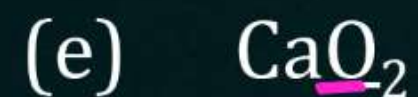
SOME SPECIAL CASES

C_3O_2 , Br_3O_8 and $\text{S}_4\text{O}_6^{2-}$

QUESTION



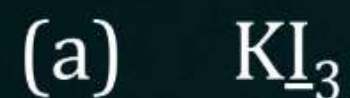
Assign oxidation number to the underlined elements in each of the following species:



Handwritten signature or mark inside an oval.

QUESTION

What are the oxidation number of the underlined elements in each of the following and how do you rationalise your results?



Ans

QUESTION



H_2S acts only as reductant where as SO_2 acts as both oxidant and reductant. Why?

[PYQ]

Homework



Sara Practice Karna.

#XUV
Thank
You

A large, hand-drawn style red heart outline is positioned to the right of the text.